2016 Intro. to Signals & Communications Exam

1. a. i. $F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-j\omega t}\, dt$

$= \int_{-b}^{b} a\, e^{-j\omega t}\, dt$ (2)

$= a \int_{-b}^{b} e^{-j\omega t}\, dt$

Some students do not know the relationship between $e^{j\theta}$ and $\sin\theta$ and $\cos\theta$

$= a \cdot \frac{1}{-j\omega} \cdot \left(e^{-j\omega b} - e^{j\omega b} \right)$

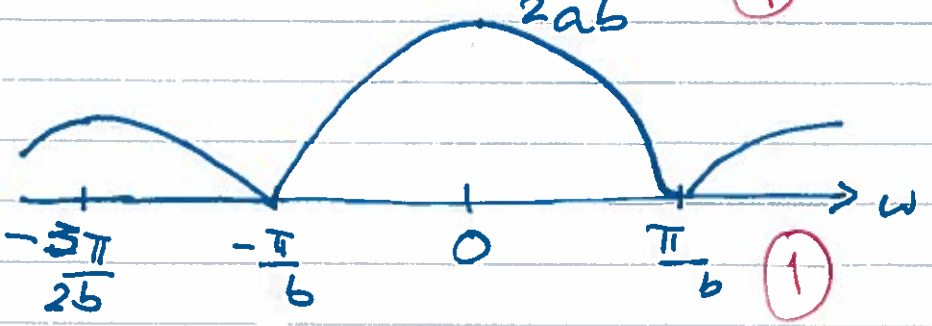
$= a \cdot \frac{1}{-j\omega} \cdot \left(-2j \sin(\omega b) \right)$

$\Rightarrow F(\omega) = \frac{2a \sin(\omega b)}{\omega}$ (2)

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$F(\omega) = 2ab \cdot \mathrm{sinc}(\omega b)$

ii) (1)

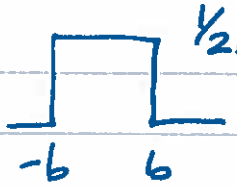


(1)

3/

phase $\angle F(\omega) = 0$ because $F(\omega)$ is real

(1)

1. a. iii. $f(t) = \lim_{b \to 0}$ 

2/

$\Rightarrow f(t) = \delta(t)$ (1) ∵ the area is unit and the width tends to zero

In this special case

$F(\omega) = \mathcal{F}[\delta(t)] = 1.$ (1)

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iv. $\mathcal{F}[f(t-t_0)]$

$= \int_{-\infty}^{\infty} f(t-t_0) e^{-j\omega t} dt$ (1)

$= \int_{-\infty}^{\infty} f(t-t_0) e^{-j\omega(t-t_0)} \cdot e^{-j\omega t_0} dt$

$= e^{-j\omega t_0} \int_{-\infty}^{\infty} f(t-t_0) e^{-j\omega(t-t_0)} dt$

$= e^{-j\omega t_0} \cdot F(\omega)$

$\Rightarrow \hat{F}(\omega) = e^{-j\omega t_0} F(\omega)$ 3/

Some students mistakenly gave the relationship as $\hat{F}(\omega) = e^{j\omega t_0} F(\omega)$

1. b. i. $\omega_0 = \frac{2\pi}{T_0}$ 1/

ii. a_0 is the dc component of the signal $g(t)$. 1/

iii. a_n reflects the degree of how $g(t)$ is similar to $\cos(n\omega_0 t)$

b_n : similarity w/ $\sin(n\omega_0 t)$ 2/

iv. If $g(t)$ is odd function of t,

$$g(-t) = -g(t) \quad (1)$$

So,

$$g(-t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(-n\omega_0 t)$$

$$= a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) - \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$

Since $g(-t) = -g(t)$

$$\Rightarrow a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) - \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$

$$= -a_0 - \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) - \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$

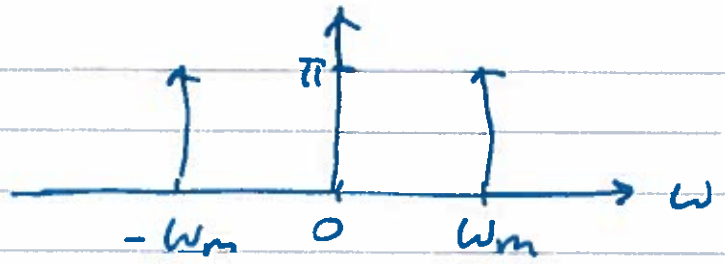
$$\Rightarrow a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) = 0 \quad (2) \quad 3/$$

$$\Rightarrow a_0 = 0, \; a_n = 0 \; \forall \text{ all } n = 1, 2, \ldots \infty$$

b_n can be of any value.

Alternative explanation is also acceptable. Based on physical intuition

1. c. i.



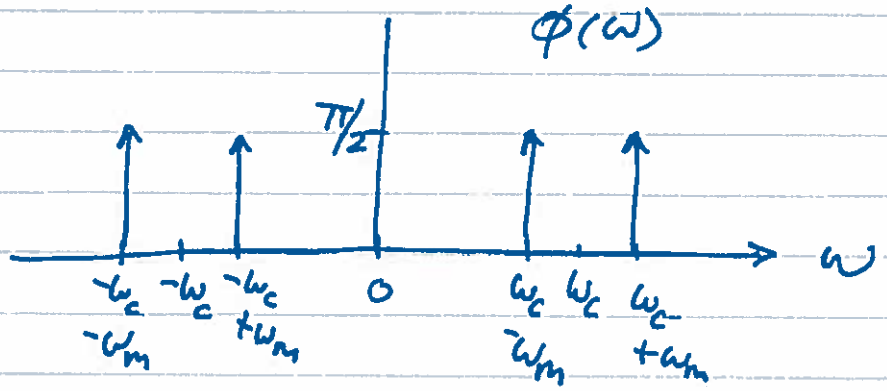
2/

ii. $\phi(t) = m(t)\cos(\omega_c t)$

$= \cos(\omega_m t).\cos(\omega_c t)$ 2/

$= \frac{1}{2}\left[\cos(\omega_c - \omega_m)t + \cos(\omega_c + \omega_m)t\right.$

iii.

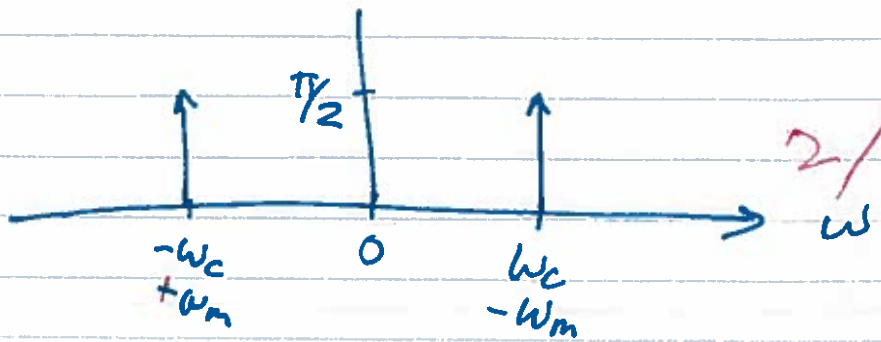


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iv. The AM effectively shifts the spectrum of $\hat{m}(t)$ to the frequency range surrounding the carrier frequency $\pm \omega_c$. 1/

v. $\phi_{USB}(t) = \frac{1}{2}\cos(\omega_c - \omega_m)t$ 2/

vi.



2/

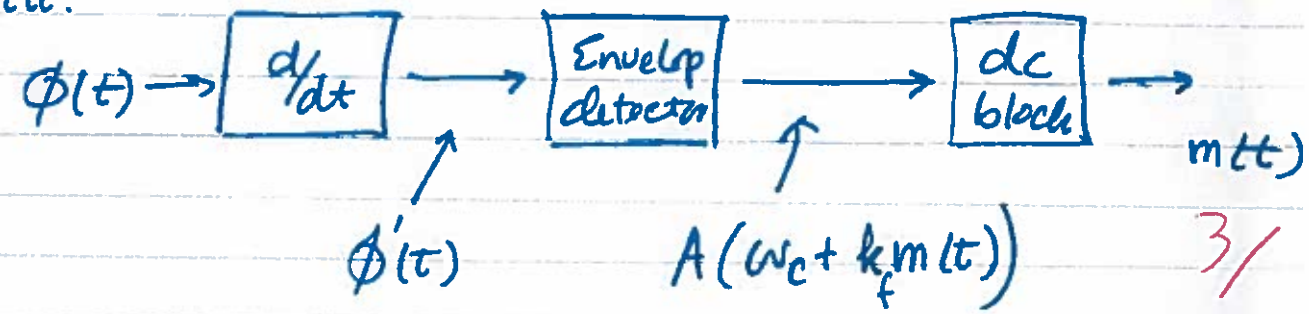
1. d. i. $\phi(t) = A\cos\left(\omega_c t + k_f \int m(\alpha)\,d\alpha\right)$ 2/

ii. $\phi'(t) = d\phi(t)/dt$

$= A\left[\omega_c + k_f m(t)\right] \cdot$

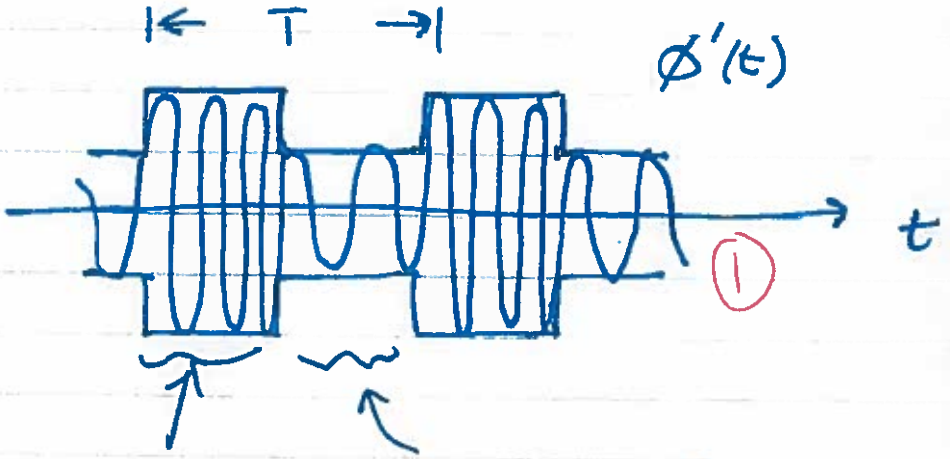
$\sin\left[\omega_c t + k_f \int m(\alpha)\,d\alpha\right]$ 2/

iii.



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iv.



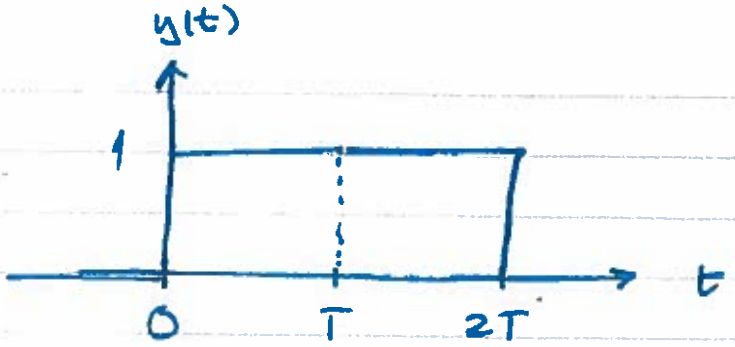
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Some students forget to give the instantaneous freq. ω_i

2. a. i.

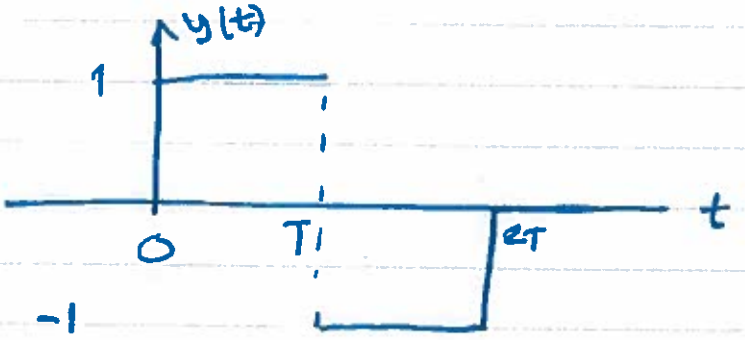
Some students forgot the definition of unit impulse response.



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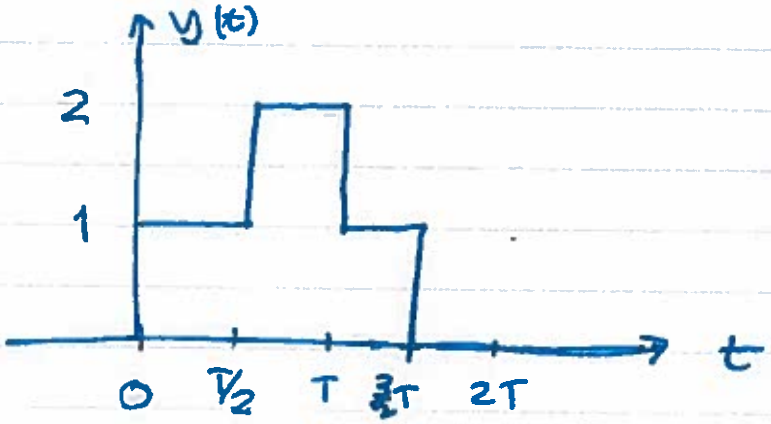
ii.

If one knows that, this question can be answered readily.



3/

iii.



3/

iv. maximum signal rate :

$$f = 1/T$$ (3)

It is so because each impulse (signal) has a response time (duration) of T time units (sec). To avoid use of complicated technique for receiving, keep the signal rate of $1/T$ samples per sec will be supported by the channel. (3)

6/

2. b. i. $f(t) * g(t)$

$$= \int_{u=-\infty}^{\infty} f(u)\, g(t-u)\, du$$

Some students forgot the definition of convolution

3/

ii. $\mathcal{F}[f(t) * g(t)]$

$$= \int_{t=-\infty}^{\infty} \int_{u=-\infty}^{\infty} f(u)\, g(t-u)\, e^{-jwt}\, du\, dt$$

(2)

$$= \int_{u=-\infty}^{\infty} \int_{t=-\infty}^{\infty} f(u)\, g(t-u)\, e^{-jwt}\, dt\, du$$

$$= \int_{u=-\infty}^{\infty} \int_{t=-\infty}^{\infty} f(u)\, g(t-u)\, e^{-jw(t-u)} \cdot e^{-jwu}\, dt\, du$$

$$= \int_{u=-\infty}^{\infty} f(u)\, e^{-jwu} \int_{t=-\infty}^{\infty} g(t-u)\, e^{-jw(t-u)}\, dt\, du$$

$$= \int_{u=-\infty}^{\infty} f(u)\, e^{-jwu} \cdot \int_{t'=-\infty}^{\infty} g(t')\, e^{-jwt'}\, dt'\, du$$

where $t' = t - u$

$$= \int_{u=-\infty}^{\infty} f(u)\, e^{-jwu}\, G(w)\, du$$

5/

$$= G(w) \cdot F(w).$$

2. b. iii. $F(\omega) * G(\omega)$

$$= \int_{u=-\infty}^{\infty} F(u)\, G(\omega - u)\, du$$

2/

iv. $\mathcal{F}[f(t)\, g(t)]$

$$= \int_{t=-\infty}^{\infty} f(t)\, g(t)\, e^{-j\omega t}\, dt$$

(2)

$$= \int_{t=-\infty}^{\infty} \frac{1}{2\pi} \int_{u=-\infty}^{\infty} F(u) e^{jut}\, du\, g(t) e^{-j\omega t}\, dt$$

$$= \frac{1}{2\pi} \int_{t=-\infty}^{\infty} \int_{u=-\infty}^{\infty} F(u)\, g(t)\, e^{-j(\omega - u)t}\, du\, dt$$

$$= \frac{1}{2\pi} \int_{u=-\infty}^{\infty} F(u) \int_{t=-\infty}^{\infty} g(t)\, e^{-j(\omega - u)t}\, dt\, du$$

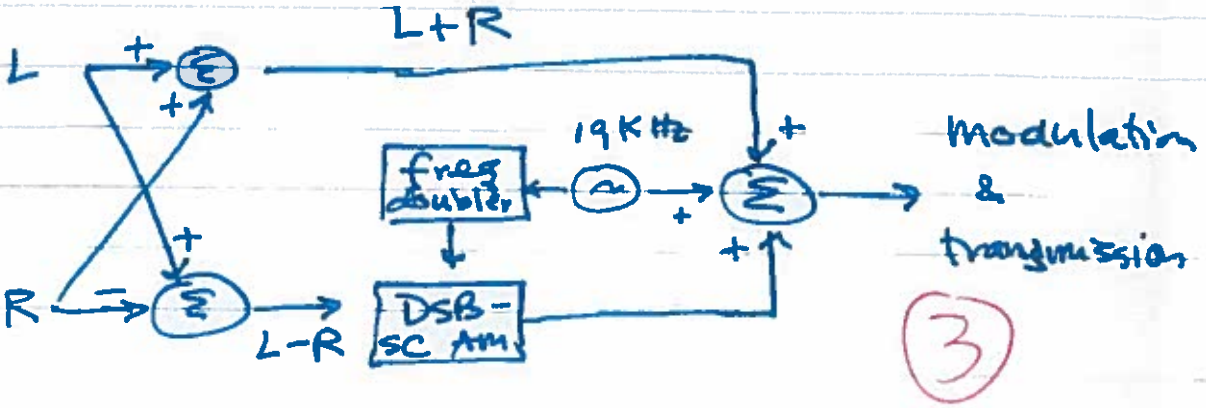
$$= \frac{1}{2\pi} \int_{u=-\infty}^{\infty} F(u) \cdot G(\omega - u)\, du$$

$$= \frac{1}{2\pi} F(\omega) * G(\omega)$$ as result in part iii. shows

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Some students were not able to derive this because of the double integral.

3.9. i.



3

Form signals : L+R and L-R.

Use & double the pilot signal of 19 KHz.

Form DSB-SC signal for L-R signal using the 38 KHz Carrier.

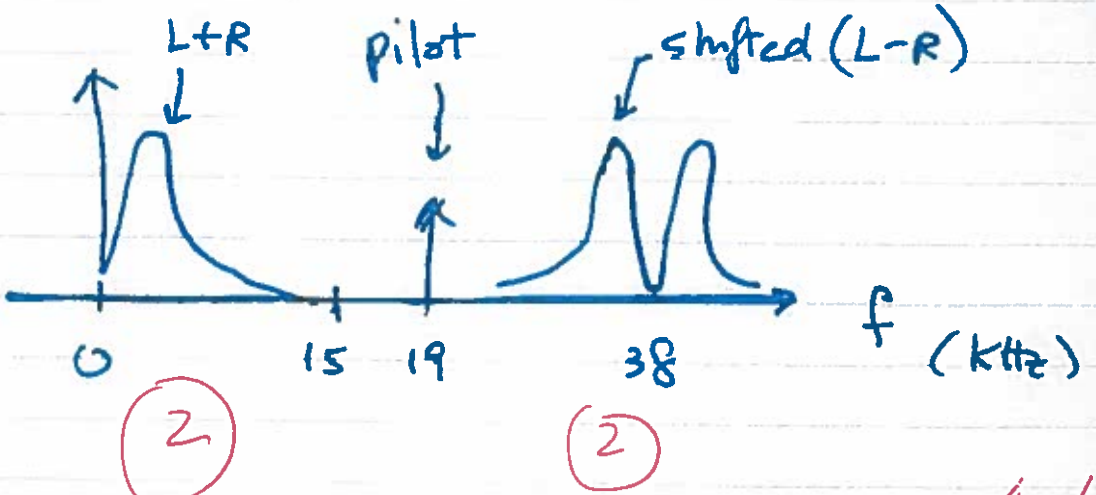
Transmit the composite base band signal of

L+R + 19 KHz + DSB-SC of L-R signal at 38 KHz Carrier

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Some students did not recall this idea of L+R & L-R channels as discussed at the lecture

ii.



2 2

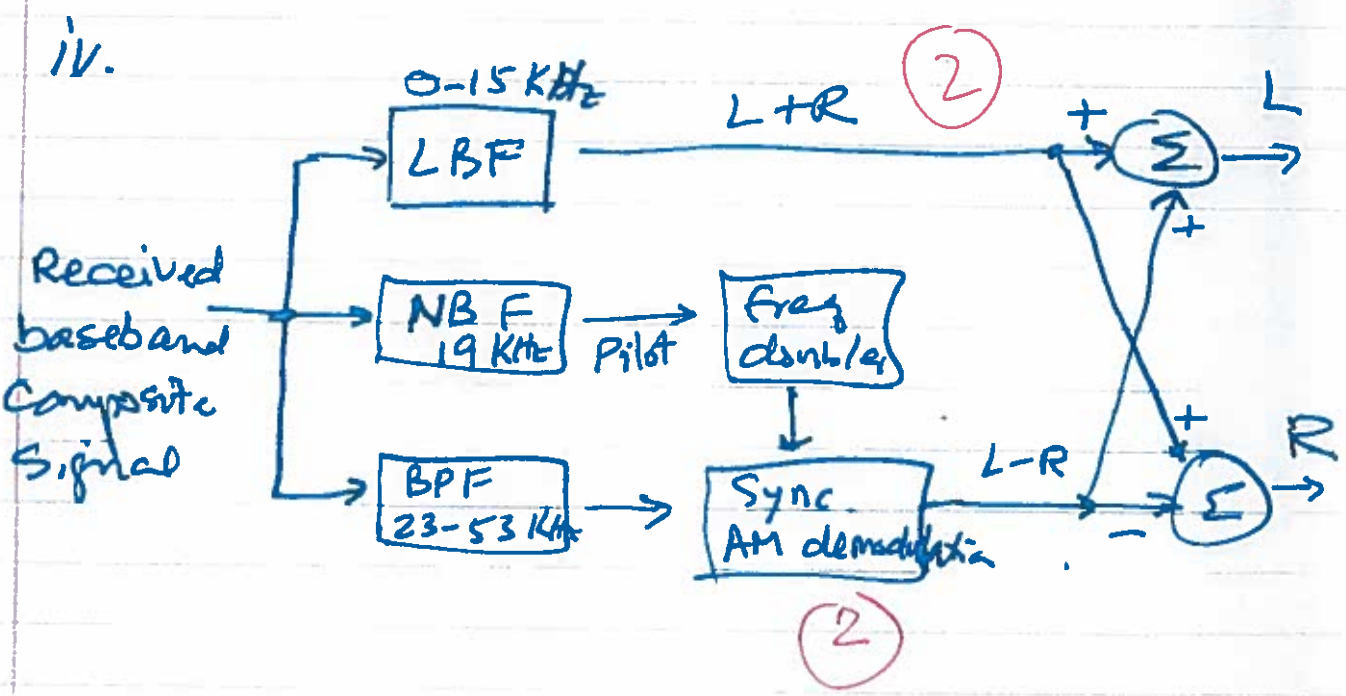
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3.a. iii. The frequency of the pilot signal of 19 KHz is doubled.

Then, the 38 KHz is multiple with the L-R signal to form the DSB-SC (AM) signal at carrier frequency of 38 KHz.

The frequency multiplication is equivalent to shift the L-R signal to the carrier frequency of 38 KHz. 4/

iv.



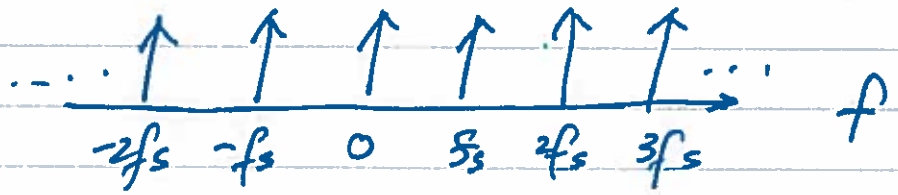
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3. b. i. Due to Nyquist Sampling Criterion:

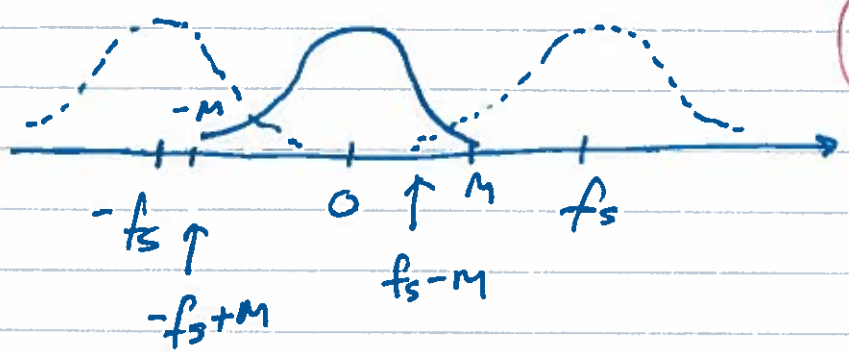
$f_s \geq 2 \times$ Bandwidth of $m(t)$ (2)

So, $M = f_s/2$

This is so because the periodic samples are equal to multiplying a train of impulses with the signal $m(t)$. The train of impulses has a spectrum of



As a result of sampling, the samples of $m(t)$ have the following frequency spectrum:



(2)

Standard question, most students could answer

From the above diagram, in order to recover $m(t)$, we require

$f_s - M \geq M$ to avoid overlap of replica of the original signal spectrum

$\Rightarrow \quad f_s \geq 2M$

6/

3. b. ii. Clearly, $C \geq M$

because the channel bandwidth must be wider than the signal bandwidth in order to avoid loss of information. 2/

iii. Since $M = f_s/2$,

we have $C \geq f_s/2$

i.e., $f_s \leq 2C$ 2/

iv. $R = 2C \log_2 M$

$= 2C \log_2 4$

$\Rightarrow R = 4C$ bits/sec 2/